

Quantum-Proof Systems Scanner

1. Executive Summary

Project Title

Quantum-Proof Systems Scanner for Public-Facing Applications

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Institution

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Domain

Cybersecurity, Applied Cryptography, Secure Systems Engineering

Overview

The Quantum-Proof Systems Scanner is a cybersecurity platform developed as part of the PNB Cyber Security Hackathon 2026. The project focuses on evaluating the cryptographic posture of public-facing applications and assessing their readiness for the transition toward Post-Quantum Cryptography (PQC).

The platform performs TLS-based security assessments of internet-facing services, extracts cryptographic metadata from digital certificates, generates a Cryptographic Bill of Materials (CBOM), evaluates cryptographic risks, and provides security analytics through a centralized dashboard.

The project was designed and implemented using Python and Django and includes features such as secure authentication with two-factor authentication (2FA), reporting capabilities, risk scoring, and enterprise-style security analytics.

This work allowed exploration of real-world cryptographic systems, TLS security mechanisms, certificate management, and emerging challenges posed by quantum computing to current public key cryptography.

Key Contributions

- Designed and implemented a TLS security scanning platform.
- Built a CBOM generation system.
- Developed a PQC readiness evaluation engine.
- Implemented risk scoring and cybersecurity analytics.

- Deployed a working web prototype on a production server.
- Studied cryptographic inventory management and quantum-safe migration concepts.

The project demonstrates practical cybersecurity engineering skills, secure software development practices, and an interest in applied cryptography and future security technologies.